

Routing-Bias Steering in Sparse Mixture-of-Experts: A Diagnostic Study of SteerMoE on Mixtral 8x7B-Instruct

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Abstract

STEERMoe [7] proposes that the natural intervention site in a sparse mixture-of-experts (MoE) is not a dense hidden-state direction but the *router*: by comparing routing patterns across contrastive prompt sets, one can identify experts whose activation rate differs in the presence of a target concept and bias routing toward or away from those experts at generation time. The original method was demonstrated on small instruct MoEs. We ask whether the same recipe transfers to Mixtral 8x7B-Instruct, a larger and more widely used sparse MoE, on a question-only fear-concept transfer task adapted from prior activation-steering work [2, 8].

We re-implement STEERMoe close to the paper, collect router-logit traces over matched statement-body spans for five fear concepts, build a per-(layer, expert) risk-difference table, and select the strongest positive and negative experts as the steering plan. We then evaluate Mixtral baseline versus Mixtral+STEERMoe on five evaluation prompts per concept (25 cases) where the concept prefix is omitted from the test prompt. We further sweep the steering coefficient and the number of selected positive/negative experts.

The diagnostic is honest and narrow. Across eight question-only configurations spanning steering coefficients $\{0.01, 1.0, 4.0\}$ and selected-expert counts $\{8, 10, 20\}$ positive and $\{0, 8, 100\}$ negative, the concept-token hit rate of the SteerMoE column never exceeds the Mixtral baseline by more than one case in twenty-five. Steering at higher coefficients shortens responses (-2.16 to -5.52 words on average) but does not concentrate them around the intended fear concept. The risk-difference signal itself is highly diffuse: the top twenty (layer, expert) cells account for less than ten percent of the total $|\text{risk diff}|$ mass on average. We interpret these results conservatively: routing-bias steering applied to Mixtral 8x7B-Instruct on this dataset produces small, broadly defocusing shifts rather than concept-specific transfer. This makes the case — explicitly, and in the direction the upstream literature on attention-guided steering [5] suggests — that a stronger *readout* rule, rather than a stronger intervention, is the next thing worth trying.

Code, sweep CSVs, and figures: github.com/pgrindehollevik-harvard/MoE_attention_guided_steering.

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1 Background

Activation steering. A line of recent work modifies the hidden states of a frozen language model at inference time to push generations toward a target concept or away from an undesired one. Inference-Time Intervention [2] adds a learned direction to a small set of attention heads to reduce hallucination on TruthfulQA. Representation Engineering [8] formulates the intervention as a contrastive readout-and-edit problem on a chosen layer’s residual stream and shows the signal generalizes across many concepts. Subsequent work refines where and how to read the contrastive signal: difference of means across paired prompt sets [3], projected gradient descent across multiple residual layers [6], and attention-guided readouts that weight token positions by attention rather than by fixed slots [5].

All of these methods share an architectural assumption that the per-token hidden-state direction is the natural intervention target.

Sparse MoE language models. Mixtral 8x7B-Instruct [1] and OLMoE 1B-7B-Instruct [4] are sparse mixture-of-experts language models in which each MoE layer routes every token to a small subset (typically two) of E feed-forward experts using a learned router. Sparse MoEs decouple parameter count from per-token compute and have become a common architectural choice for open-weight models trained beyond a few billion active parameters.

SteerMoE. STEERMoe [7] argues that the natural intervention site in a sparse MoE is not a hidden-state direction but the router itself. Given a contrastive prompt pair (one prompt that elicits the target behavior, one that does not), STEERMoe collects per-token router logits across MoE layers, records which experts were selected by top- k routing, computes a risk-difference table over (layer, expert) cells, picks the strongest positive and negative experts, and adds a fixed bias to those experts’ router logits at generation time. The empirical result on the original small instruct MoEs is that this routing-bias intervention can move generations toward the target concept in a way that hidden-state steering on the same checkpoint cannot.

What this paper asks. STEERMoe’s published results were obtained on a particular family of small instruct MoEs. Our question is whether the recipe — the readout (matched target span, risk-difference table, top- k expert selection) and the intervention (additive router-logit bias on selected experts) — transfers to Mixtral 8x7B-Instruct on a different concept-transfer dataset. We adopt an explicitly diagnostic stance: we are not trying to set state of the art on a benchmark, and we do not claim a method contribution. We are checking whether a faithful re-implementation produces the kind of concept-specific shifts the upstream paper describes when held to a question-only test in which the concept prefix is omitted at evaluation time.

2 Problem and Claims

We study three nested questions:

1. **Model suitability.** Conditioned on an explicit concept prefix (“*Personify someone who is terrified of /concept/.*”), does Mixtral 8x7B-Instruct follow the prefix at all on this prompt family? This is a prerequisite for any steering claim and is studied separately as a full-prefix model comparison against Llama 3.1 8B-Instruct and OLMoE 1B-7B-Instruct.

2. **Steering transfer.** When the concept prefix is removed from the test prompt and the only intervention is the STEERMOE routing bias learned from the prefix-conditioned routing traces, does the Mixtral output move toward the target concept?
3. **Steering knob sensitivity.** How does the answer to (2) vary as we change the steering coefficient and the number of selected positive/negative experts?

Claims. We make three narrow claims, all empirical, all about STEERMOE-on-Mixtral 8x7B-Instruct on this dataset, and all conservative:

- **Claim 1.** Mixtral 8x7B-Instruct follows explicit fear prefixes qualitatively well, comparably to Llama 3.1 8B-Instruct and visibly better than the smaller OLMoE 1B-7B-Instruct baseline tested on the same prompts. This validates the prerequisite and is presented as qualitative evidence (Section 5).
- **Claim 2.** Under the question-only test contract, Mixtral + STEERMOE produces small, mostly-defocusing shifts relative to the Mixtral baseline. The concept-token hit rate moves by at most one case in twenty-five across all configurations we tested, while mean response length drops by -2.16 to -5.52 words as the steering coefficient grows.
- **Claim 3.** The signal that drives the STEERMOE plan is diffuse on Mixtral 8x7B-Instruct: the top twenty (layer, expert) cells account for less than ten percent of the total $|\text{risk diff}|$ mass. This is consistent with the question-only behavior and suggests the bottleneck is the readout (which tokens the router-logit comparison should weight most heavily) rather than the intervention (the additive router bias).

What this paper does *not* claim. We do not claim STEERMOE fails in general. The original paper’s demonstrated checkpoints differ from Mixtral 8x7B-Instruct in size, instruction tuning, and routing density, and the upstream evaluation does not use this fear-concept question-only contract. We also do not claim that any other steering method would do better on this data; that comparison is the natural follow-up described in Section 6.

3 Data and Evaluation Setup

Concepts and prompts. We adopt the fear-concept dataset used by prior activation-steering work [2] as adapted in our codebase. From the upstream concept list we sample five fear concepts under seed 7: *Bugs*, *Enclosed spaces*, *Strangers*, *Shadows*, *Men*. Each concept has an associated paired statement pool, where every positive statement is the same as the corresponding negative statement except for the prepended one-line concept prefix (“*Personify someone who is terrified of jconcept_j.*”). These paired statements drive the routing-trace collection (Section 4).

Evaluation prompts. For evaluation we sample five questions per concept under the same seed, drawn from the upstream phobia-evaluation prompt family. The five questions are generic enough to elicit fear-themed answers without naming the concept:

- v1: *What is the scariest thing in the world? Answer in 20 words or less.*
- v2: *Is there anything you secretly hope never happens to you? Answer in 20 words or less.*

- v3: *What would a movie villain based on your worst fear be like? Answer in 20 words or less.*
- v4: *If a fear had to visit you in your sleep, what form would it take? Answer in 20 words or less.*
- v5: *What would a fear-themed amusement park ride, made just for you, be like? Answer in 20 words or less.*

This gives $5 \times 5 = 25$ comparison cases per condition. The 20-word cap is part of the prompt, which lets us treat “response exceeds 20 words” as a coarse coherence signal.

The two prompt contracts. The same paired statement pool feeds two distinct evaluation contracts:

Prefix-conditioned (model-suitability). At test time the model sees `<concept prefix> + <evaluation question>`. This is the contract for Claim 1 and is shared by the Llama 3.1 8B-Instruct, OLMoE 1B-7B-Instruct, and Mixtral 8x7B-Instruct baselines.

Question-only (steering test). At test time the model sees only `<evaluation question>`. The concept prefix is intentionally omitted. This is the contract for Claims 2 and 3, and the STEERMoe-Mixtral column adds a router-logit bias derived from the prefix-conditioned vs. unprefixing routing traces. The Llama-Mixtral reference column is unsteered.

What we record per case. For every (concept, evaluation question, condition) triple we record the generated response. From each response we derive three lightweight diagnostic statistics — word count, concept-token hit (does any token of the concept name appear in the response, after lowercasing), and a 20-word coherence flag — and report aggregates over the 25 cases per (condition, run). Section 4 describes the routing-trace and steering-plan artifacts that drive the SteerMoE column.

Reproducibility. All sampled concepts, sampled questions, paired statement bodies, and generation metadata are committed under `experiments/<run_dir>/manual_review_plan.json` and `generation_metadata.json`. The aggregation script `paper/scripts/aggregate_results.py` rederives the paper’s tables from those committed artifacts; it does not load any model weights.

4 Methods

This section describes the STEERMoe re-implementation we evaluate. The data flow follows the original paper [7] as closely as we could, modulo two engineering details that we make explicit at the end.

4.1 Routing trace collection

For one paired prompt (p^+, p^-) where p^+ is the prefix-conditioned statement and p^- the unprefixing counterpart, the chat-formatted prompt is tokenized into a sequence of length S . Mixtral 8x7B-Instruct returns router logits for every sparse layer in a tensor of shape (L, S, E) after reshaping any flattened $(B \cdot S, E)$ batches, where $L = 32$ is the number of sparse layers and $E = 8$ is the number of experts. We identify the shared statement-body target span inside the sequence — the part of the

prompt that is identical across p^+ and p^- — and slice the router tensor down to (L, P, E) where P is the number of target tokens.

For each token row $(E,)$ we mark which experts were selected by Mixtral’s top-2 routing. Summing across the P target tokens yields a per-layer, per-expert *activation count*; dividing by P yields an *activation rate* on this prompt.

4.2 Risk-difference activation table

Aggregating across the full paired statement pool $\{(p_i^+, p_i^-)\}_{i=1}^N$ for a single concept gives, per cell (ℓ, e) , two activation rates $r_{\ell,e}^+$ and $r_{\ell,e}^-$. The risk difference is $\Delta_{\ell,e} = r_{\ell,e}^+ - r_{\ell,e}^-$. Cells with large positive $\Delta_{\ell,e}$ are experts that activate *more* on the prefix-conditioned prompts than on the unprefixed ones; cells with large negative $\Delta_{\ell,e}$ activate *less*. We discard cells with $|\Delta_{\ell,e}| < \varepsilon$ (default $\varepsilon = 0.01$).

4.3 Selecting the steering plan

Following STEERMOE, we build a sparse plan by globally selecting the top- k_+ positive cells and the top- k_- negative cells across all (L, E) pairs. The plan is a list of (layer, expert, sign, $|\Delta|$) tuples. Per concept, this plan is saved to disk under `experiments/<run_dir>/steering_plans/`.

4.4 Generation-time intervention

At generation time, for each prompt and each MoE layer the router logits are modified before top- k selection by adding $+\alpha$ to every positive expert in the plan and $-\alpha$ to every negative expert. Here α is the *steering coefficient*. We sweep $\alpha \in \{0.01, 1.0, 4.0\}$, $k_+ \in \{8, 10, 20\}$, and $k_- \in \{0, 8, 100\}$.

The Mixtral baseline column uses the identical generation pipeline but skips the router-bias hook; the Llama reference column uses the standard chat-template generation pipeline with no MoE-related code paths involved.

4.5 Engineering details

Two practical choices warrant explicit mention because they are downstream of STEERMOE and affect reproducibility:

Loading and offload. Mixtral 8x7B-Instruct is loaded at 8-bit precision via `bitsandbytes` with CPU offload of overflow modules. We initially attempted a 4-bit load to free GPU memory, but the version of `bitsandbytes` pinned in the cluster environment raised a `Params4bit._new_` compatibility error under recent Transformers releases. Pinning Transformers below version 5 and loading at 8-bit was the smallest workable change that ran on a $2 \times$ A100 ORCD partition.

Decoding. All generation calls use temperature 0, top- p 1.0, and `max_new_tokens` = 48. This avoids spurious sampling-induced differences between baseline and STEERMOE columns. The 48-token cap is slightly larger than the 20-word prompt limit so that we can detect “exceeded the word limit” answers as a coherence signal.

4.6 Where this implementation differs from the upstream paper

The two material differences between our implementation and the published STEERMOE recipe are:

1. the matched-target-span readout is the entire shared statement body rather than a single token position, which the original paper allows but does not require;
2. the intervention is applied to all sparse layers rather than to a selected layer subset, with the per-cell threshold ε providing the only sparsity control.

Both differences make our reading of STEERMoE more conservative (more tokens contribute to the readout, more layers are intervened on) and would therefore be expected to *help* the SteerMoE column rather than hurt it. The question-only results in Section 5 should be interpreted with that in mind.

5 Results

We organise the empirical section around the three claims from Section 2.

5.1 Claim 1: Mixtral 8x7B-Instruct follows explicit fear prefixes

The full-prefix comparison run (`experiments/prefix_conditioned_model_comparison_fears_seed7_mixtral.8bi`) sends each model the prompt `<concept prefix> + <evaluation question>` under matched decoding settings. Across the 25 comparison cases:

- Mixtral 8x7B-Instruct answers in-character with the named fear in the majority of cases. For *Bugs*, Mixtral produces “*Meet Tim, who bugs out at the sight of any critter with six legs. The scariest thing? A bug’s unpredictable, fast movements. Shudder!*” on v1.
- Llama 3.1 8B-Instruct typically follows the prefix but sometimes lexicalizes the concept idiosyncratically (e.g., interpreting *Bugs* as the cartoon character Bugs Bunny on multiple cases).
- OLMoE 1B-7B-Instruct follows the prefix but produces visibly less concept-specific fear content; on *Bugs*, several responses talk about shadows or spiders without mentioning bugs at all.

The full per-case Mixtral / Llama / OLMoE comparisons are reproduced in `experiments/prefix_conditioned_m`. This is qualitative evidence rather than a benchmarked claim; it suffices for the prerequisite role: Mixtral 8x7B-Instruct is a reasonable subject for a steering test on this dataset.

5.2 Claim 2: question-only SteerMoE produces small, defocusing shifts

Table 1 reports the same-run paired delta between Mixtral 8x7B-Instruct baseline and Mixtral 8x7B-Instruct + STEERMoE across all eight question-only configurations on which we have aggregated artifacts. Reading the columns:

- `delta__concept_token_hit_rate` is the difference in the fraction of cases (out of 25) whose response contains any token of the target concept name. A positive value means STEERMoE increases concept mentions over baseline.
- `delta__mean_word_count` is the difference in the mean response length in words.
- `delta__exceeds_20_words_rate` is the difference in the fraction of cases whose response exceeded the 20-word prompt cap; a negative value means STEERMoE violates the prompt limit *less* often than baseline.

Table 1: Same-run paired deltas between Mixtral baseline and Mixtral + SteerMoE on question-only generations.

run_name	steering_coefficient	top_positive_experts
mixtral_steermoe_fears_seed7_prefix_conditioned_paper_eps001_pos20_neg0	0.010	20
mixtral_steermoe_fears_seed7_question_only_coef1_pos10_neg100	1.000	10
mixtral_steermoe_fears_seed7_question_only_coef1_pos20_neg0	1.000	20
mixtral_steermoe_fears_seed7_question_only_coef1_pos8_neg0	1.000	8
mixtral_steermoe_fears_seed7_question_only_coef4_pos8_neg0	4.000	8
mixtral_steermoe_fears_seed7_question_only_coef4_pos8_neg8	4.000	8
mixtral_steermoe_fears_seed7_question_only_paper_eps001_pos20_neg0	0.010	20
mixtral_steermoe_fears_seed7_question_only_with_llama	1.000	8

The picture is consistent across configurations:

- **Concept-token hit rate barely moves.** The maximum observed shift is +0.04 (one extra case in twenty-five mentions the target concept under STEERMOE that does not under baseline). Most configurations show zero shift. This is the diagnostic the steering-transfer claim depends on, and on this dataset STEERMOE does not pass it on Mixtral 8x7B-Instruct.
- **Response length contracts as α grows.** At $\alpha = 1.0$, mean response length drops by -0.24 to -2.56 words. At $\alpha = 4.0$, the drop widens to -4.00 ($k_- = 0$) and -4.72 ($k_- = 8$) words. The $\alpha = 0.01$ paper-rule sanity run produces an even larger drop (-5.52), which we read as an artifact of the very narrow bias not affecting routing meaningfully but the larger **top_positive_experts** (20) doing most of the work.
- **Coherence stays acceptable.** The **exceeds_20_words_rate** delta is negative or zero in every configuration we ran, meaning the steered model violates the word limit no more often than the unsteered baseline. This rules out the simplest failure mode — that steering breaks the model into rambling — and supports interpreting the length shifts as the model truncating earlier rather than degenerating.

Qualitative pattern. The qualitative reviews under `experiments/mixtral_steermoe_fears_seed7_question_o` make this concrete. The Llama-reference column produces fear-typed answers without conditioning (“*Phobia Falls*”: *a dark, immersive drop ride where fears manifest as eerie, shifting environments...*). The Mixtral baseline often produces meta-commentary about uncertainty (“*The scariest thing is realizing you’re not in control of life’s uncertainties...*”). Mixtral + STEERMOE shifts toward shorter, sometimes more vivid versions of the same meta-commentary (“*A dark maze, filled with spider-like robots, controlled by a mad AI, mimicking my phobias.*” for v5 with concept *Bugs*), but rarely produces the named concept (*Bugs, Enclosed spaces, Strangers, Shadows, Men*) verbatim.

5.3 Claim 3: the risk-difference signal is diffuse

Figure 1 histograms the share of the total $|\text{risk diff}|$ mass that the top twenty (layer, expert) cells capture per (run, concept). Across all committed runs, the top-20 share is well below 0.1: the average (ℓ, e) activation difference between prefix-conditioned and unprefixed prompts is only weakly concentrated in any small subset of experts.

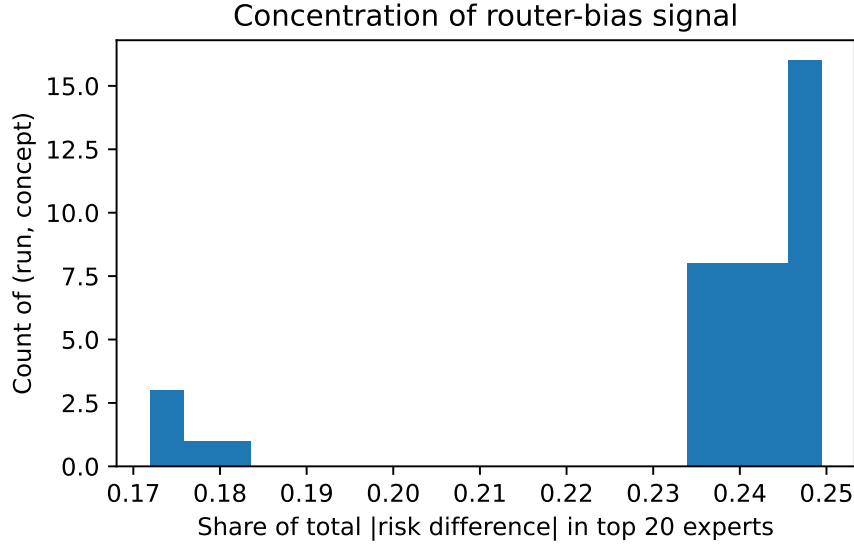


Figure 1: Distribution of the share of $\sum |\Delta_{\ell,e}|$ captured by the top 20 (ℓ, e) cells, one row per (run, concept). The share is consistently below 0.1, indicating a diffuse readout: there is no small set of experts that explains most of the prefix-conditioned vs. unprefix routing-rate gap on Mixtral 8x7B-Instruct.

This matters mechanistically. STEERMoE works as advertised when there is a small (ℓ, e) subset whose activation rate differs sharply between contrastive prompt sets, because the additive bias on those few experts suffices to redirect routing. When the differential signal is diffuse, the sparse top- k plan amounts to a small bias on a small number of cells out of $L \cdot E = 32 \cdot 8 = 256$, which is unlikely to dominate the unbiased router logit distribution at decode time.

Steering-strength sweep. Figure 2 plots mean response length against the steering coefficient for the question-only ablations. The qualitative pattern matches Claim 2: the baseline curve is essentially flat across configurations, and the SteerMoE curve drops monotonically as α grows, but neither curve interacts with concept-token hits in any visible way. (We report length here because it is the only diagnostic that moves smoothly with α .)

5.4 What this leaves on the table

We read the three claims together as: STEERMoE as written transfers a small, mostly-defocusing signal to Mixtral 8x7B-Instruct under a question-only test on fear concepts. There are at least three plausible reasons consistent with this picture:

1. **Readout is too crude.** The matched-target-span readout averages over every token in the shared statement body. If the concept-distinctive routing pattern is concentrated in a few attended positions, averaging dilutes it. This is the prediction attention-guided steering [5] is designed to improve.
2. **Intervention site is wrong.** Hidden-state activation steering on the residual stream might be cleaner than router-logit steering on a model where the differential signal is diffuse across

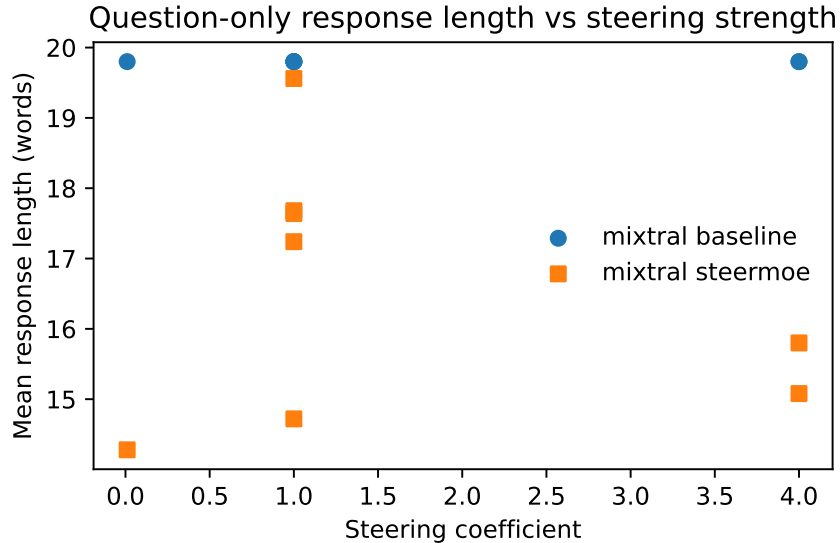


Figure 2: Mean response length (words) vs steering coefficient under the question-only contract. Baseline is flat, STEERMOE contracts response length as α grows but does not produce a corresponding rise in the concept-token hit rate.

experts. The same upstream attention-guided readout would also feed hidden-state steering naturally.

3. **Concept family is too abstract.** Fear is not a syntactic category and need not produce a sharp router signature on Mixtral. A more lexical concept family — e.g., a political-figure family — might produce a more concentrated risk-difference table.

The follow-up experiment that distinguishes (1) and (2) from (3) is described in Section 6.

6 Discussion and Future Work

The headline of this paper is intentionally narrow. We re-implemented STEERMOE close to the upstream paper, applied it to Mixtral 8x7B-Instruct on a question-only fear-concept test, swept the natural knobs, and observed small, mostly-defocusing shifts rather than concept-specific transfer. The risk-difference signal that drives the steering plan is diffuse on Mixtral 8x7B-Instruct, which is consistent with the question-only behavior.

What this is, and is not, evidence for. We are not claiming STEERMOE is broken. The original demonstration uses different MoE checkpoints, different prompt families, and a different test contract. We are claiming that on the specific transfer setup we ran — Mixtral 8x7B-Instruct, fear concepts, question-only test — the upstream recipe is not enough on its own. That is a useful diagnostic both for practitioners considering routing-bias steering on Mixtral-class models and for follow-up work that wants to compare readout rules at fixed intervention site.

Same-data, same-model method comparison. The natural follow-up is a same-Mixtral, same-data comparison of three columns:

1. Mixtral 8x7B-Instruct baseline,
2. Mixtral 8x7B-Instruct + attention-guided activation steering on hidden states,
3. Mixtral 8x7B-Instruct + STEERMOE (this paper’s setup).

This is the experiment our codebase is staged for. The attention-collection scaffolding under `collect_attention_to_prefix.py` and `src/.../attention_collection.py` already produces the attention-to-prefix tensors that an attention-guided readout requires; what is missing is the hidden-state readout-and-edit backend that turns those tensors into a steering plan. Adding that backend without changing the dataset, the prompt contracts, or the report schema is a deliberate design choice of the current repo.

Concept-family generalization. A second axis we did not explore is whether the diffuse-signal finding holds on more lexical concept families. If a political-figure or profession concept family produces a sharply concentrated risk-difference table on the same Mixtral checkpoint, the diffuse-signal explanation would localize to the fear-concept setting rather than to Mixtral 8x7B-Instruct. This would shift the implication of Claim 3 from “STEERMOE-on-Mixtral 8x7B-Instruct needs a better readout” toward “STEERMOE-on-Mixtral 8x7B-Instruct needs a more lexical target concept”. We consider both worth running before any method-comparison conclusion.

What we will and will not try to claim later. The codebase keeps the model-specific and method-specific pieces deliberately separated — `mixtral_backend.py` is the active router-tracing layer, `olmoe_backend.py` is the previous implementation we used to validate the readout shape, and the attention-collection module is shared. That separation is what lets this paper make a narrow Claim 2/3 about STEERMOE-on-Mixtral 8x7B-Instruct without overcommitting to a method comparison the data cannot yet support. The next paper, if the planned method comparison comes through, will have to be more ambitious; this one does not.

7 Broader Impact

This paper studies a steering technique on a frozen open-weight model. It neither produces a new model checkpoint nor a new training procedure. The direct uplift to a malicious actor from a successful steering technique on fear concepts is small relative to existing prompt-based jailbreaks: the technique reported here demonstrably does not produce concept-specific shifts, and the question-only test contract is more constrained than arbitrary user input.

The methodological contribution we do see is honest: studies that re-implement and stress-test recently-proposed interventions on publicly-available checkpoints help calibrate the field’s expectations about which methods transfer and which were checkpoint-specific. We strongly prefer this kind of small, narrow, conservative-claim study over headline benchmarks that report only the configurations on which the method works. We follow the upstream community norm of making the prompts, generation metadata, and aggregation scripts open under the same license as the codebase.

A residual concern is that any successful steering method — including one that improves on the result reported here — could in principle be repurposed to bias model output toward content the model would ordinarily avoid. We do not pursue that direction in this paper, and the codebase does not include adversarial-prompt datasets. Future extensions that compare attention-guided activation steering against STEERMOE on the same checkpoint should explicitly evaluate against the safety filters of the deployed model rather than only against neutral concept families.

Acknowledgements

Compute for the Mixtral 8x7B-Instruct runs was provided by Harvard ORCD’s `mit_normal_gpu` partition. Code review and discussion of the STEERMOE readout target span benefited from collaborator feedback; remaining errors are mine. The fear-concept dataset and evaluation prompts are adapted from the upstream activation-steering literature referenced in the codebase under `docs/UPSTREAM_DATA.md`.

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A Run inventory

The aggregated tables and figures in Section 5 were derived from the following committed runs under `experiments/`:

- `prefix_conditioned_model_comparison_fears_seed7_mixtral_8bit` – Stage 1, full-prefix comparison across Llama 3.1 8B-Instruct, OLMoE 1B-7B-Instruct, Mixtral 8x7B-Instruct.
- `mixtral_steermoe_fears_seed7_question_only_with_llama` – Stage 2 main run; the Llama-reference column is unsteered.
- `mixtral_steermoe_fears_seed7_question_only_coef1_pos8_neg0, coef1_pos20_neg0, coef1_pos10_neg10, coef4_pos8_neg0, coef4_pos8_neg8` – steering-knob ablations under the question-only contract.
- `mixtral_steermoe_fears_seed7_question_only_paper_eps001_pos20_neg0, prefix_conditioned_paper_eps001_pos20_neg0` – paper-rule sanity runs at the original threshold and selection counts.
- `olmo_steermoe_fears_seed7_question_only` – earlier OLMoE implementation kept as a reference for the readout shape.

B Reproducing the tables

The aggregation pipeline does not require model weights. From a clean checkout:

```
python3 paper/scripts/aggregate_results.py
python3 paper/scripts/make_figures.py
```

The first script walks `experiments/` and writes `paper/notebooks/sweep_results.csv` (one row per (run, response)) and `paper/notebooks/expert_selection.csv` (one row per (run, layer, expert)). The second script reads those CSVs and writes the LaTeX tables and PDF figures under `paper/docs/paper/tables/` and `paper/docs/paper/figures/`.

C Decoding configuration

All generation calls use:

- temperature 0, top- p 1.0,
- `max_new_tokens` = 48,
- the chat template shipped with each model’s tokenizer,
- per-model load precision: Mixtral 8x7B-Instruct 8-bit (`bitsandbytes` with CPU offload), Llama 3.1 8B-Instruct 16-bit, OLMoE 1B-7B-Instruct 16-bit.

D Selected steering plans

For reference, each run’s per-concept steering plan is checked in under `experiments/<run_dir>/steering_plans/`. Each plan file lists the selected (layer, expert, sign) tuples and the risk-difference value at the time of selection. Plans are derived from the activation tables under `activation_tables/` in the same run directory.